Range–IMU Localization under Degenerate Geometry

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1. Motivation and Contribution

Motivation

- Estimators often assume good anchor geometry without assessing it.
- Under weak geometry, noise and outliers can drive estimates towards an incorrect solution branch.

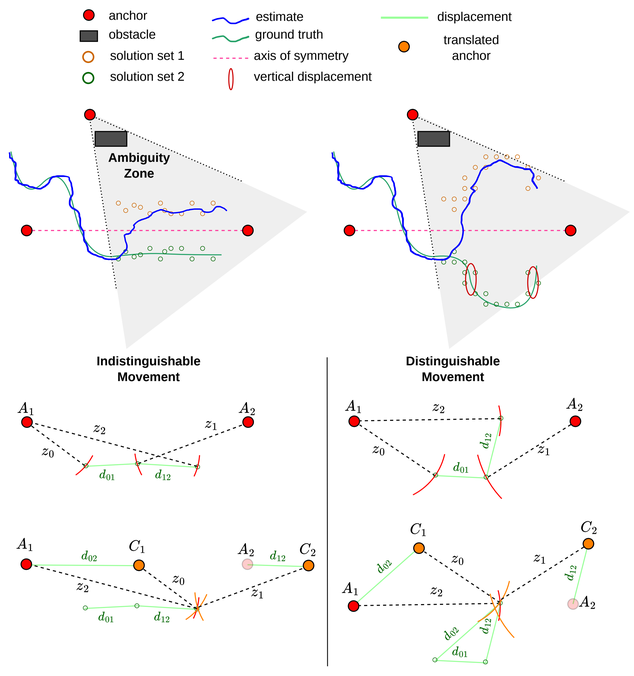
Contribution

- Geometry-aware framework for weak anchor constellations.
- Two terminal-position solvers: Geo-ML (maximum likelihood) and Geo-QCLS (quadratically constrained least squares).
- Fusion that reduces corrections along weakly observable directions.

2. Translated-anchor geometry

Translate past anchors by estimated displacement $d_{k|T}$ [1,3] to constrain terminal position x_T:

$$c_{k|T} = a_i + d_{k|T}, \qquad z_k \simeq \|x_T - c_{k|T}\|.$$



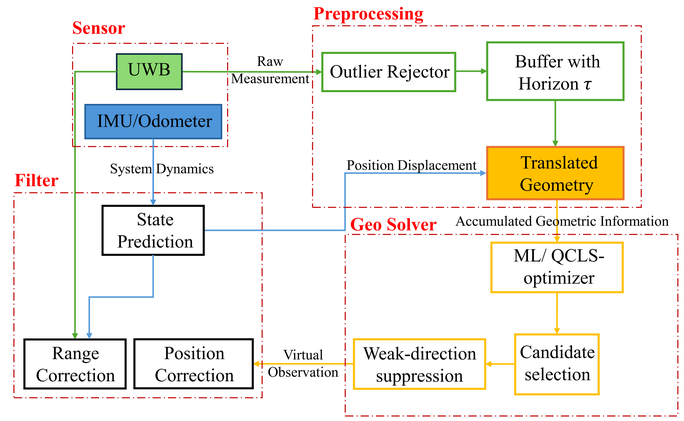
Motion can preserve ambiguity or improve geometry.

Let κ be the centres' affine dimension and n_p the position dimension. Then, for consistent, noise-free ToA ranges [4], we typically have:

$$\begin{cases} \kappa < n_p - 1, & \text{infinitely many solutions,} \\ \kappa = n_p - 1, & \text{at most two symmetric solutions,} \\ \kappa = n_p, & \text{a unique solution.} \end{cases}$$

In 3-D, $\kappa = 1, 2, 3$ means collinear, coplanar and noncoplanar centres.

3. Position solvers



Framework: geometry-aware position recovery and fusion.

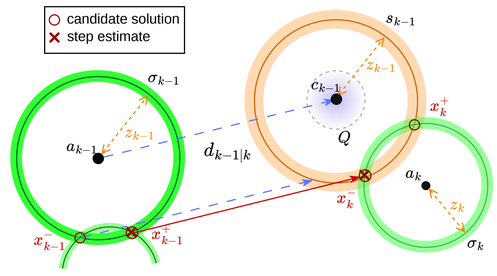
- **Geo-ML.** Iteratively reweight an approximate likelihood using:

$$s_k(x) = \sigma_k^2 + g_k^\top Q_d(k) g_k.$$

- **Geo-QCLS.** Refine ranges $q_i^\star$ with measurements and motion; solve:

$$\min_{x,\eta} \sum_i w_i \big(\eta - 2c_i^\top x + \|c_i\|^2 - \rho_i\big)^2,$$
$$\eta = x^\top x, \qquad \rho_i = (q_i^\star)^2.$$

Both solvers recover **only the terminal position**.



Temporal candidate selection.

- **Candidate selection.** When multiple plausible solutions exist, we need a selection strategy. For example, use translated geometry at two adjacent time steps to select candidates based on displacement consistency.

4. Geometry-aware fusion

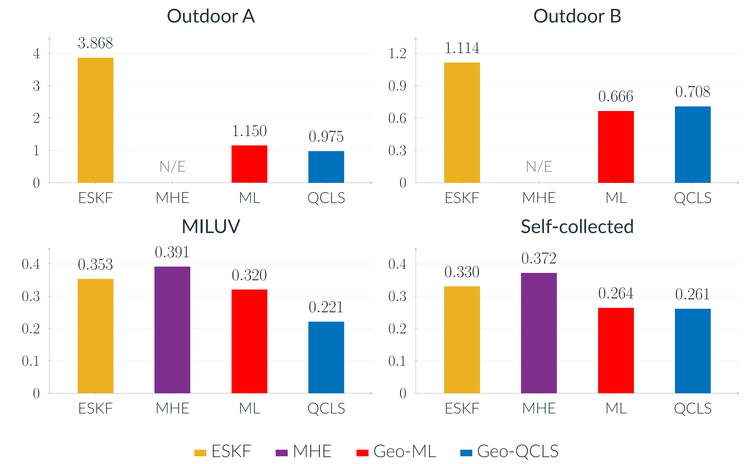
Directional information is obtained from the range Jacobians:

$$I = G^\top S^{-1} G = V \operatorname{diag}(\lambda_j) V^\top.$$

- **Identify weak directions.** Small relative eigenvalues reveal directions poorly constrained by the accumulated ranges.
- **Suppress corrections.** Inflate covariance along weak directions to reduce their correction strength.
- **Fuse position.** Use the recovered position as a virtual measurement.

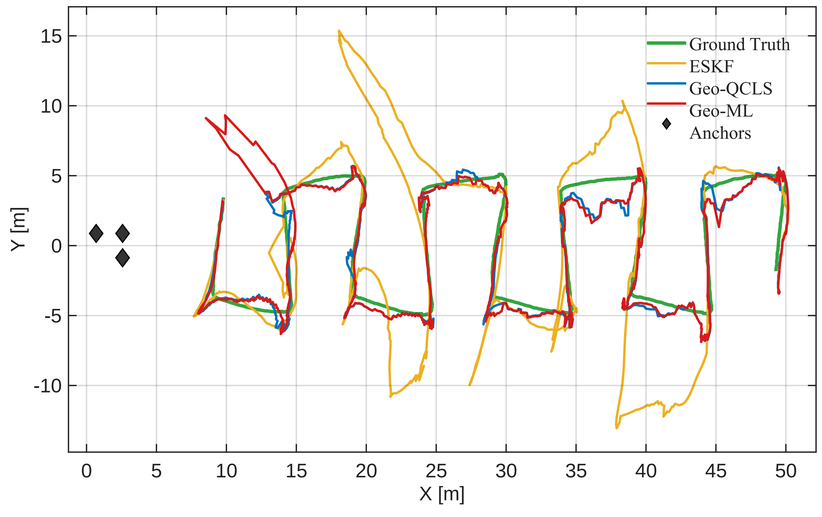
5. Experiment and Conclusion

- Tests: outdoor 2-D [5], MILUV indoor 3-D [6], and self-collected flights with sequential anchor dropouts.
- Benchmarks: ESKF and 3-D moving horizon estimation (MHE).



RMSE: root mean square error (m).

Geo-QCLS reduces RMSE versus ESKF by 74.8%, 37.4% and 20.9% on Outdoor A, MILUV and self-collected data, respectively.



Outdoor A: clustered anchors and estimated trajectories.

Translated-geometry adds no information beyond the recorded ranges and motion, but reveals the weak directions and guides how much to trust the current information in each direction. By combining historical ranging measurements with estimated motion, the proposed framework recovers a virtual position measurement and suppresses corrections along weak geometric directions.

Selected references

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